

Language-Guided Temporal Token Pruning for Efficient VideoLLM Processing

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Abstract

Vision Language Models (VLMs) face significant computational challenges with long-form videos due to attention mechanisms’ quadratic complexity. We propose Language-Guided Temporal Token Pruning (LGTTP), which extracts temporal cues from queries to adaptively assign pruning rates across video frames, preserving contextual continuity while reducing computational overhead. Unlike existing methods that apply uniform token pruning or disruptive keyframe selection, LGTTP maintains higher token density in temporally relevant segments. Our model-agnostic approach integrates with TimeChat and LLaVA-Video architectures, demonstrating substantial efficiency gains across diverse benchmarks. Experiments show LGTTP reduces computation by 65% while preserving 97-99% of original performance. On highlight detection, LGTTP achieves +9.5% HIT@1 on QVHighlights compared to baseline methods, and for temporal grounding, it maintains 99.6% of original R@1 on Charades-STA. LGTTP shows particular strength for queries with explicit temporal markers while remaining effective for general video understanding tasks.

1 Introduction

Vision Language Models like TimeChat (Ren et al., 2024), LLaVA-Video (Zhang et al., 2024), and VideoLLaVA (Lin et al., 2023) have demonstrated exceptional capabilities across various video understanding tasks, from highlight detection to temporal grounding and video question answering. However, these models face substantial computational inefficiency when processing long-form videos, as they typically encode every frame into visual tokens and process the entire token sequence for each query. The computational complexity grows quadratically with sequence length due to the attention mechanism, making efficient token management a critical challenge for practical

deployment. Real-world video queries often target specific temporal segments, making full sequence processing inefficient.

Current efficiency approaches fall into two categories with significant limitations: (1) vision token pruning methods like PruMerge (Shang et al., 2024) and ToMe (Bolya et al., 2023), which reduce spatial redundancy within individual frames but fail to capture temporal connections; and (2) keyframe selection methods like KeyVideoLLM (Liang et al., 2024) and VideoTree (Wang et al., 2024), which disrupt temporal context by completely discarding intermediate frames. These limitations are particularly problematic for temporal understanding tasks that rely on maintaining temporal coherence across frames, such as highlight detection and temporal grounding.

We propose Language-Guided Temporal Token Pruning (LGTTP), a model-agnostic approach that addresses these limitations by adaptively assigning pruning rates based on temporal cues extracted from queries. LGTTP integrates effectively with both TimeChat and LLaVA-Video architectures, showing particular strength with models that have built-in temporal awareness. By leveraging temporal elements, LGTTP preserves tokens most relevant to temporal queries while reducing computational requirements by 65% and maintaining 97-99% of original performance.

Specifically, our contributions include: (i) A model-agnostic framework that extracts temporal cues from natural language queries to guide token pruning decisions. (ii) A method to integrate LGTTP with TimeChat and LLaVA-Video architectures, with particular optimization for temporally-aware models. (iii) Comprehensive evaluation across multiple video understanding benchmarks, demonstrating that LGTTP maintains near-original performance while significantly reducing computational requirements for diverse tasks.

2 Method

Overview and Motivation. LGTTP addresses a fundamental efficiency challenge in VideoLLMs: not all frames are equally relevant to a given query, particularly for temporal understanding tasks. While existing approaches either prune tokens uniformly or select entire keyframes, LGTTP adaptively preserves the most temporally relevant tokens while maintaining contextual continuity. Our approach consists of three components: temporal cue extraction from queries, temporally-aware relevance prediction for different models, and adaptive token pruning based on predicted relevance. By leveraging temporal indicators in queries, we can concentrate computational resources on the most relevant segments.

Temporal Cue Extraction. Natural language queries often contain rich temporal information that guides our pruning strategy. Given a query Q , we extract temporal information through two stages: First, we identify temporal markers (e.g., “before,” “after,” “during”) using pattern matching and a fine-tuned classifier, categorizing them into *Precedence*, *Subsequence*, or *Co-occurrence* relationships. Second, we extract reference events and their potential temporal positions - for example, from “after talking to the coach,” we identify “talking to the coach” as occurring earlier in the video. This temporal knowledge helps prioritize frames likely to contain moments of interest.

Adaptation to VideoLLM Architectures. LGTTP integrates with various VideoLLM architectures according to their temporal awareness capabilities. For timestamp-aware models like TimeChat, we leverage existing timestamp bindings where frame embeddings $E = \{e_1, \dots, e_N\}$ already incorporate timestamp information. For models using temporal instructions (e.g., LLaVA-Video), we add lightweight temporal position embeddings based on normalized frame positions. For standard VLMs without explicit temporal awareness, we introduce a temporal adapter that projects frame indices into a positional embedding space.

After adaptation, we compute relevance scores by processing the query embedding $e_q = F_{text}(Q)$ and calculating base relevance $L_{base} = a \cdot \cos_sim(E', e_q) + b$, where a and b are learnable parameters. We then incorporate extracted temporal cues through a weighting mechanism $L_{temp} = L_{base} \odot W_{temp}$, where W_{temp} prioritizes frames

based on identified temporal relationships.

Temporally-Adaptive Token Pruning. The final step converts temporal relevance scores into frame-specific pruning rates. The frame-wise pruning rates are computed as:

$$R = (r_1, r_2, \dots, r_N) = \alpha N \cdot \text{softmax}(L_{temp}) \quad (1)$$

where α controls the overall pruning rate and N is the frame count. This ensures the average pruning rate across frames is approximately α , while distribution varies based on temporal relevance.

Critically, we employ soft selection rather than hard keyframe selection. For each frame, we retain:

$$T_i = \max(T_{min}, \lceil (1 - r_i) \cdot T_{full} \rceil) \quad (2)$$

tokens, where T_{min} is the minimum token count (typically 10% of the original count T_{full}). This preserves contextual continuity by maintaining some tokens even from less relevant frames.

3 Experiments

3.1 Experimental Setup

Datasets: We evaluate LGTTP across multiple video understanding benchmarks: QVHighlights (Lei et al.) for moment retrieval tasks with human-written natural language queries; Charades-STA (Gao et al., 2017) for temporal grounding that evaluates the ability to locate specific activities described in text; VideoMME (Fu et al., 2024) for comprehensive video question answering that tests general understanding capabilities across diverse scenarios; and EgoSchema (Zhang et al., 2023) for egocentric video understanding with narrative-style queries capturing first-person perspectives.

Metrics: We use task-specific evaluation metrics aligned with standard benchmarks. For highlight detection on QVHighlights, we report mAP and HIT@1. For temporal grounding on Charades-STA, we measure R@1 at IoU thresholds of 0.5 and 0.7. For video question answering on VideoMME and egocentric understanding on EgoSchema, we report accuracy on multiple-choice questions. Across all experiments, we measure computational efficiency using FLOPs relative to the original unmodified models.

Model Architectures: We implement LGTTP with two complementary VideoLLM architectures: (1) TimeChat (Ren et al., 2024), which features a timestamp-aware frame encoder ideal for temporal tasks, and (2) LLaVA-Video (Zhang et al.,

Method	FLOPs	Highlight Detection QVHighlights		Temporal Grounding Charades-STA	
		mAP	HIT@1	R@1 (IoU=0.5)	R@1 (IoU=0.7)
TimeChat (original)	100%	21.7%	37.9%	46.7%	23.7%
Random Sampling	35%	14.2%	25.6%	34.5%	15.8%
ToMe	38%	15.5%	27.3%	36.2%	16.9%
PruMerge	35%	16.3%	28.9%	37.8%	17.6%
KeyVideoLLM	40%	13.1%	27.0%	32.1%	14.5%
KVTP	35%	19.6%	34.2%	42.3%	21.2%
LGTPP (Ours)	35%	21.2%	43.7%	46.5%	23.1%

Table 1: Performance comparison on highlight detection (QVHighlights) and temporal grounding (Charades-STA) tasks with TimeChat at 65% pruning rate.

2024) in both 7B and 13B parameter configurations, representing standard architecture with temporal instruction prompts. This selection allows us to evaluate LGTPP’s effectiveness across models with varying temporal awareness capabilities and parameter scales.

Baselines: We compare LGTPP against: Original unmodified models. Random token sampling, Uniform pruning methods: PruMerge (Shang et al., 2024) and ToMe (Bolya et al., 2023), Keyframe selection: KeyVideoLLM (Liang et al., 2024), KVTP (Liu et al., 2025): A recent state-of-the-art approach for video token pruning.

3.2 Results on Video Understanding Tasks

Performance on Temporal Tasks. Table 1 shows LGTPP’s performance on highlight detection (QVHighlights) and temporal grounding (Charades-STA) when integrated with TimeChat. Our approach maintains near-original performance (within 0.5-0.8%) despite reducing computation by 65%. On QVHighlights, LGTPP achieves 21.2% mAP and 43.7% HIT@1, significantly outperforming other efficiency methods, including KVTP (34.2% HIT@1). For temporal grounding on Charades-STA, LGTPP achieves 46.5% R@1 at IoU=0.5, matching the original model’s performance while using only 35% of the computation. This demonstrates LGTPP’s particular strength in preserving critical temporal relationships when pruning tokens.

Performance on General Video Understanding. Table 2 shows LGTPP’s effectiveness when integrated with LLaVA-Video on question answering and egocentric understanding tasks. Notably, LGTPP maintains performance within 0.6-1.1% of the original models across both 7B and 13B variants. On VideoMME, LGTPP achieves 62.0% accuracy with the 7B model, which is effectively on par with the unpruned model (62.6%). This in-

dicates that LGTPP’s temporal awareness benefits extend beyond explicit temporal tasks to general video understanding, where maintaining context across frames remains important.

Cross-Model Analysis. Our cross-architecture experiments reveal that LGTPP provides consistent benefits regardless of the base model, though with varying magnitudes. The largest improvements come when integrated with TimeChat (+9.5% HIT@1 on QVHighlights compared to KVTP), likely because TimeChat’s timestamp-aware architecture provides more precise temporal information for LGTPP to leverage. With LLaVA-Video, gains are more modest but still significant (+0.8% on VideoMME), showing LGTPP’s effectiveness even with standard temporal modeling approaches.

Query-Dependent Performance. A key observation is that LGTPP’s advantage varies with query type. For queries with explicit temporal markers like “before/after,” LGTPP outperforms KVTP by 7.2% HIT@1 on QVHighlights. For “during/while” markers, the advantage is 5.8%, while for queries without explicit temporal markers, it maintains a 2.3% improvement. This gradation confirms our hypothesis that language-guided pruning is most beneficial when temporal relationships are explicitly expressed, though it provides benefits across all query types.

Efficiency-Performance Balance. Across all experiments, LGTPP achieves an optimal efficiency-performance balance. On highlight detection, it retains 97.7% of the original mAP performance while on temporal grounding it maintains 99.6% of R@1 (IoU=0.5) performance. For video QA tasks, it preserves 99.0% of the accuracy, all while reducing computation by 65%. This makes LGTPP particularly suitable for real-world deployment scenarios where computational resources are constrained.

Method	FLOPs	Video Question Answering VideoMME		Egocentric Understanding EgoSchema	
		7B	13B	7B	13B
LLaVA-Video (original)	100%	62.6%	69.5%	54.2%	65.8%
Random Sampling	35%	58.3%	62.4%	50.7%	60.5%
ToMe	38%	58.9%	62.9%	51.5%	61.2%
PruMerge	35%	59.8%	64.5%	52.5%	63.1%
KeyVideoLLM	40%	51.3%	60.5%	46.8%	55.2%
KVTP	35%	61.8%	66.3%	52.4%	63.6%
LGTP (Ours)	35%	62.0%	67.1%	53.1%	64.0%

Table 2: Performance on video question answering (VideoMME) and egocentric understanding (EgoSchema) tasks with LLaVA-Video at 65% pruning rate.

3.3 Ablation Studies

Temporal Cue Extraction Impact. Removing the temporal cue extraction component significantly degrades performance, with varying impact across tasks. For highlight detection, we observe a -6.5% reduction in HIT@1 on QVHighlights, while temporal grounding on Charades-STA drops by -5.1% in R@1. Even general video QA tasks see a -1.8% reduction in accuracy, demonstrating that temporal understanding benefits all video tasks to some degree.

Architectural Integration Strategies. Our comparison of different integration approaches reveals that adaptation strategy matters. The lightweight temporal adapter approach outperforms simple position embedding by 2.1% on QVHighlights and 1.5% on VideoMME. This suggests that effectively modeling temporal relationships requires more sophisticated integration than just positional information, especially for models without built-in temporal awareness.

Soft vs. Hard Selection Effects. The difference between soft and hard selection is particularly pronounced. When using hard selection (pruning rate = 0 or 1), performance drops significantly across all tasks: -9.3% HIT@1 on QVHighlights, -7.6% R@1 on Charades-STA, and -3.5% accuracy on VideoMME. These results strongly validate our design choice to maintain a minimum token threshold for all frames, preserving contextual continuity that proves crucial for video understanding.

Pruning Method Integration. LGTP improves all baseline pruning methods, but the magnitude varies. When combined with PruMerge, we see +14.8% HIT@1 improvement on QVHighlights and +8.7% R@1 on Charades-STA. With ToMe, the improvements are +16.4% and +10.3% respectively. This demonstrates LGTP’s versatility as a framework that can enhance existing token reduction methods by adding temporal awareness to

their pruning decisions.

4 Limitations

Despite LGTP’s effectiveness, it has several limitations: (1) Performance depends on the presence of temporal cues in queries, with reduced benefits for queries lacking explicit temporal markers; (2) Our implementation handles basic temporal relationships but struggles with complex reasoning involving multiple temporal constraints; (3) While the preprocessing overhead is minimal (0.3-0.5% of total inference time), it requires additional computational steps; and (4) Optimal integration requires architecture-specific adaptations, potentially limiting straightforward deployment across all VideoLLM variants.

5 Conclusion

We presented Language-Guided Temporal Token Pruning (LGTP), which leverages temporal cues in queries to adaptively prune visual tokens in VideoLLMs. By preserving temporally relevant tokens while maintaining contextual continuity, LGTP reduces computation by 65% with minimal performance loss across highlight detection, temporal grounding, and video QA tasks. Our approach demonstrates consistent advantages over existing efficiency methods, particularly for temporally complex queries, preserving 97-99% of original performance while achieving up to +9.5% HIT@1 improvements over baseline pruning methods. LGTP enables more practical deployment of powerful VideoLLMs for long-form video analysis, making advanced video understanding more accessible in resource-constrained environments and real-time applications. Future work will focus on handling more complex temporal relationships and exploring extensions to emerging multimodal architectures.

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A Appendix

B Related Work

B.1 Video Language Models

Recent advances in video language modeling have produced a variety of architectures with different approaches to temporal understanding. TimeChat (Ren et al., 2024) incorporates explicit timestamp awareness by binding visual content with corresponding timestamps, enabling accurate temporal localization. LLaVA-Video (Zhang et al., 2024) uses a temporal instruction preprompt to inform the LLM about the video sampling process. VideoLLaVA (Lin et al., 2023; Cheng et al., 2024) employs a learnable temporal embedding within the vision encoder. While these models demonstrate strong capabilities, they face computational challenges when processing long videos, particularly for tasks requiring precise temporal understanding. Other approaches like VideoChat (Li et al., 2023), Vid2Seq (Yang et al., 2023), and Video-ChatGPT (Maaz et al., 2024) have also shown promising results in video understanding tasks but typically process frames independently, making them less optimal for capturing fine-grained temporal relationships.

B.2 Efficiency Approaches for VLMs

Several approaches have been proposed to address the computational challenges of vision token processing in VLMs. Vision token pruning methods like PruMerge (Shang et al., 2024), ToMe (Bolya et al., 2023), FastV (Chen et al., 2024), Vid-TLDR (Choi et al., 2024), and DynamicViT (Rao et al., 2021) focus on reducing token counts based on importance metrics like attention entropy or attention scores. However, these methods typically operate on individual frames without considering temporal context. Keyframe selection methods like KeyVideoLLM (Liang et al., 2024), VideoTree (Wang et al., 2024), and Koala (Tan et al., 2024) identify and retain only the most informative frames, but their hard selection approach disrupts temporal coherence. KVTP (Liu et al., 2025) bridges these approaches by using soft selection to retain some tokens from less relevant frames, but it doesn’t explicitly leverage temporal cues from queries.

B.3 Video Understanding Benchmarks

The evaluation of VideoLLMs spans multiple specialized benchmarks. For moment retrieval, QVHighlights (Lei et al.) and Charades-STA (Gao et al., 2017) are commonly used to evaluate a model’s ability to locate temporal segments matching natural language queries. General video understanding capabilities are assessed on benchmarks like VideoMME (Fu et al., 2024) and EgoSchema (Zhang et al., 2023), which include diverse question-answering tasks. Step localization is evaluated on datasets like NeXT-QA (Xiao et al., 2021) and COIN (Yansong Tang, 2019). Our approach aims to improve efficiency across this diverse range of tasks without compromising performance.

C Implementation Details

We implement LGTTP by integrating it with TimeChat and LLaVA-Video pipelines. For base token pruning, we adopt PruMerge due to its effectiveness in preserving important visual information, though our approach is compatible with other pruning methods. The computational overhead of our complete pipeline is minimal, adding only 0.3-0.5% to overall inference time while reducing computation by 65%.

D Temporal Marker Classification

We develop our temporal marker classifier without manual annotation using a weakly supervised approach. First, we construct a comprehensive lexicon of temporal markers by extracting common temporal expressions from existing NLP resources (e.g., “before,” “after,” “during,” “while,” “when”). We then use rule-based pattern matching to automatically identify and classify these markers in 5,000 queries from QVHighlights, Charades-STA, and VideoMME training sets.

The automatically labeled dataset is used to train a 2-layer MLP classifier on top of frozen BERT embeddings. To improve robustness, we employ data augmentation techniques including synonym replacement and word reordering to create variations of queries with preserved temporal relationships. We also include queries without explicit markers as negative examples.

Our classifier achieves 88.6% accuracy when evaluated against a small manually verified test set of 200 queries. For queries without explicit markers but containing implicit temporal references

(e.g., “show the beginning”), we rely on a predefined vocabulary of temporal terms and their relative positions in the video timeline. This approach allows us to effectively identify temporal relationships without requiring extensive manual annotation.